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ReActive - smart pants for functional electrical stimulation (FES) and EMG monitoring

Project Report

Sport Engineering - Industrial Design Studio for Wearable Sport

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Abstract: This report presents the concept and design of a smart garment integrating surface electromyography (EMG) and Functional Electrical Stimulation (FES), developed to enhance the autonomy of elderly users in the context of the Active^3 – Everyone,Everywhere,Everyday project. Designed to work in synergy with the FES-bike developed at Politecnico di Milano, the trousers aim to simplify electrode placement, reduce wiring, and improve usability during assisted cycling. The development followed a structured design process, starting from the definition of target users and system requirements, supported by literature review, patent analysis, and market research. The solution addresses the current lack of garments that integrate both EMG and FES technologies. Electrodes placement was based on SENIAM guidelines, using bipolar EMG electrodes (∅10mm, 20mm electrodes spacing) and rectangular FES electrodes (4×8cm) with rounded edges to optimize comfort and stimulation efficiency. Positions were defined over the gluteus maximus, rectus femoris, biceps femoris, and gastrocnemius medialis. Manual and digital mapping was performed using size L Decathlon leggings, and a sewing pattern was created in Autodesk Inventor. The garment integrates Nanoleq's GECKO ElectroSkin electrodes, PhantomLink connectors, and PhantomTape X conductive traces, ensuring stretch resistance, washability, and low skin-electrode impedance (<150 kΩ), with minimal impact on breathability. The system architecture separates the FES and EMG subsystems: FES electrodes connect to the preexisting trike stimulator, while custom-designed EMG units with integrated IMUs and wireless transmission handle biosignal acquisition. User interfaces include color-coded snap-fit connectors and intuitive placement markers. A preliminary prototype validated EMG signal acquisition on the rectus femoris, confirming feasibility and guiding future improvements. The proposed solution offers a scalable, wearable platform for combined EMG-FES rehabilitation tailored to the needs of elderly users.

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1. Introduction

In the panorama of the project Active^3 – Everyone,Everywhere,Everyday [1], sport is considered a key way to promote inclusion and therapy through assisted physical activity. An example is the development of the FES-Bike: a trike embedded with a system for functional electrical stimulation and an electric engine to assist users during cycling. The device is used for several purposes, mainly in the Cybathlon event in which athletes that do not have control over their lower limbs compete with the assistance of the functional electrical stimulation.
Another purpose of the trike, of interest for the actual project, is to promote sport for inclusion and active ageing. Physical activity can typically induce improvements in muscle health avoiding its atrophization and in cardiovascular health. This effects are determinant for elderly people that have to face decrease of muscle mass, balance and neuromuscular functions while ageing. Indeed in the project Active^3 – Everyone,Everywhere,Everyday the trike aims to promote assisted cycling (through electrical stimulation and engine) to elderly subjects that struggle with exercise, also due to physical impairments or pain.
The trike is composed by a main frame to which several components are attached: an electric engine to assist the cycling activity; an encoder that monitors the crank angle to stimulate muscles at the correct timing; sensorised pedals with the addition of leg orthosis to monitor the force and keep the legs fixed; a stimulator to activate the electrodes; a Raspberry control software to link the encoder with the stimulator, and eventually a battery to power all the electronics (Figure 1). Moreover, the trike is integrated with specific buttons to change the stimulation strategy depending on the cycling cadence and with an application that can manage more specifically the stimulation parameters.
At the moment, the electrodes actually used for the stimulation require to be placed in the correct position by an expert person and connected to the stimulator for an efficient and safe activity. This limitation can reduce the will of elderly subjects to perform the exercise autonomously in a future and possible scenario.
For this reason, a focus group with six participants (age over 65 years old) of the project Active^3 – Everyone,Everywhere,Everyday that tried the trike at least once was held to find possible improvements related to any aspect of the device. The focus group was held outside of the actual project for external and different purposes. However, the main issue that was highlighted relied on the lack of autonomy: specifically the getting on/off phase from the trike and the placement of the electrodes on the correct muscles showed strong agreement between the participants. As previously mentioned, this last problem is crucial for the independent use and should be solved with an intuitive and safe alternative.
In this panorama the current project aims to design smart pants with integrated electrodes for functional electrical stimulation that can be connected directly to the trike's stimulator for a fast and efficient use. The main objective is to promote autonomy and ease of use for elderly subjects that participate in the project Active^3 - Everyone,Everywhere,Everyday. The smart garment should also embed EMG electrodes to record muscle activity and fatigue during the volitional activity (not in parallel with electrical stimulation) in order to monitor the beneficial effects over multiple times of exercise.
In the following sections, all the steps that led to the ideation of a design concept of the pants will be deeply explained. The design process that was followed includes the definition of the brief, the research activity to analyze the context and the requirements, the ideation phase and a preliminary prototype.

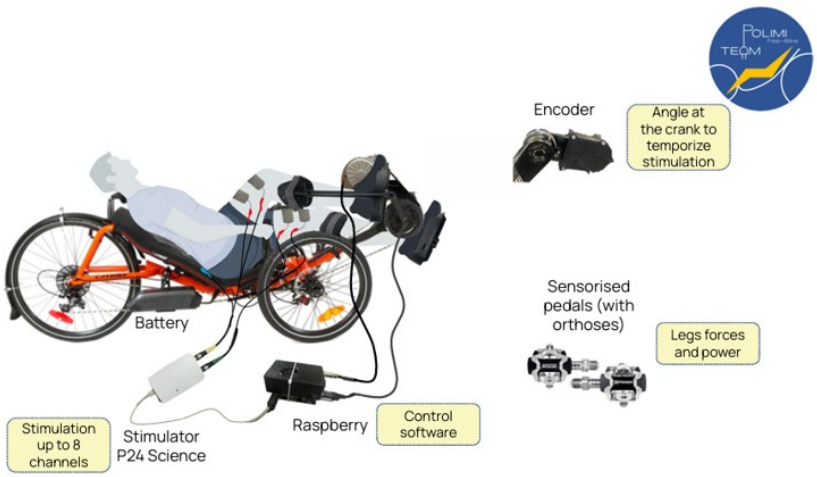

Figure 1: *scheme and main components of the Polimi FES-Bike.*

2. Brief Definition

The first step of the design process was the formulation of the brief. It was necessary to clearly define the objective of the project, the target users and the overall context of the product to then proceed with a straightforward idea of the project's possibilities. For this reason, a research activity was also performed to analyze the sources of information related to the topic of smart garments used for FES and EMG monitoring. This included a review of the literature and the patents, and an analysis of the current market.
Eventually, by having a broader view on the context of interest, a list of the product's requirements was compiled.

2.1. Objective

The objective of the design process (up to proof-of-concept) relies on the development of a smart garment integrated with electrodes for functional electrical stimulation (FES) of the muscles of the lower limbs and for EMG monitoring of the muscles' activity and fatigue. The two functions should work at different moments: the EMG monitoring is thought for trainings in which the electrical stimulation is not used and the biological working principle of the muscles is activated (this permits to record the possible improvements and long-term beneficial effects of the FES). Moreover, the EMG function should be independent from the electrical stimulation channel and provided with an autonomous control unit for the wireless transmission of the data to a remote device or the data storage for offline analysis.
The product should be tailored to work specifically in the context presented above, ensuring ease of connectivity with the trike's stimulator and correct stimulation/monitoring of the muscles' activity. Eventually the product should guarantee effective level of autonomy in the placement of the electrodes respect to the actual condition, considering as final users elderly people of the project Active^3 - Everyone,Everywhere,Everyday.

2.2. Target Users

The target users for the proposed product can be identified at first as the subjects who took part into the project Active^3 - Everyone,Everywhere,Everyday related to the use of the FES-Bike for promoting active ageing. Secondly, also future users who will take advantage of the actual project that is still in progress could be considered as valuable audience. The decision to target such specific users brings with it important considerations: the wearability of the garment must be immediate and avoid misundertandings/misplacement of the elctrodes to guarantee ease of use and mainly safety. The position of the electrodes and the muscles to stimulate must be chosen to not harm the users that differently from athletes or performing subjects requires other strategies. Eventually the independency of use must be the key objective for the type of target users chosen as previously

highlighted. To better analyze the tasks the target user should perform with the new product, the main steps to the activity's approach are reported below:

FES – ASSISTED TRAINING:

1. Preparation of the electrodes for the use (soak electrodes with electrolyte if the smart garment embed wet electrodes);
2. Wearing of the smart garment, controlling the correct way of positioning;
3. Checking on the correct positioning of the electrodes (possibility to add set of instructions for the correct use of the garment);
4. Sitting on the trike;
5. Connection of the smart garment with the trike's stimulator cables;
6. Activation of the electronic devices of the trike and start of the physical activity/exercise;
7. Finish of the session and detachment of the stimulator cables;
8. Taking off the smart garment;
9. Possible washing of the smart garment and ironing after the physical activity.

EMG – AUTONOMOUS TRAINING:

1. Preparation of the electrodes for the use (soak electrodes with electrolyte if the smart garment embed wet electrodes);
2. Wearing of the smart garment, controlling the correct way of positioning;
3. Checking on the correct positioning of the electrodes (possibility to add set of instructions for the correct use of the garment);
4. Connection of the control units with the smart garment;
5. Wireless connection of the control unit with remote device;
6. Sitting on the trike;
7. Start of the EMG recording;
8. Activity session;
9. Finish of the session and of the recording;
10. Taking off the smart garment and detachment of the control units;
11. Possible washing of the smart garment and ironing after the physical activity;
12. Possible analysis of the data / sending of the data to analyst (doctor, researcher. . .)

2.3. Research

2.3.1. Literature Review & State of the Art

To properly design the product, an in-depth review of the literature on the present technologies for electrical stimulation and EMG monitoring was carried out. The research focused mostly on textile and wearable solutions, but also on stimulation control strategies, muscular activity and fatigue recognition. The following section presents the most outstanding five of the many reviewed documents, together with their key topics and results.

1. *"Ink-based textile electrodes for wearable Functional Electrical Stimulation: a proof-of-concept study to evaluate their comfort and efficacy", Master Degree Thesis of Valentina Oliveri, A.Y. 2022-23* [2]

   The Master's thesis by Valentina Oliveri (A.Y. 2022–23) presents a proof-of-concept study focused on the development and assessment of ink-based textile electrodes for Functional Electrical Stimulation (FES). The study investigates their effectiveness and comfort when compared to conventional hydrogel electrodes, highlighting the potential of textile-based solutions for wearable and reusable applications.

   The electrodes were integrated into cotton bands using conductive ink-printing techniques, with the goal of ensuring good skin contact, durability, and seamless integration into garments. The central aim was

to evaluate the feasibility of these textile electrodes in delivering electrical stimulation in a safe, efficient, and comfortable way.

Electrical performance was assessed through impedance spectroscopy. As expected, dry textile electrodes showed higher impedance than standard self-adhesive hydrogel electrodes, primarily due to the lack of a conductive gel interface. However, this increase in impedance did not prevent the textile electrodes from successfully delivering the required stimulation parameters to induce muscle contractions. The results suggest that textile electrodes can provide adequate current delivery for FES without requiring substantially higher voltages.

To evaluate stimulation performance, tests were conducted on healthy subjects involving quadriceps stimulation during knee flexion-extension movements. Indicators such as minimum stimulation threshold, range of motion, movement speed, and onset of fatigue were measured. No statistically significant differences were found between the textile and hydrogel electrodes in these parameters, indicating that textile electrodes are equally capable of inducing effective and controlled movements.

Comfort during stimulation was also examined through a qualitative questionnaire assessing discomfort and pain. Both types of electrodes were generally well tolerated, with no major differences in user-reported discomfort. However, it was noted that the textile electrodes required slight moistening (using hand cream) to ensure optimal conductivity, which may introduce variability in stimulation effectiveness and user experience.

Overall, while textile electrodes exhibit slightly lower efficiency in current transfer due to higher impedance, their effectiveness in inducing movement and maintaining user comfort remains comparable to that of hydrogel electrodes. An added benefit is their inherent potential for washing and reuse, which presents a substantial advantage in terms of sustainability and long-term usability.

The study also explored the use of electromyography (EMG) to assess muscle fatigue during stimulation. Although some recording errors limited the reliability of the EMG data in this case, the thesis highlights EMG analysis as a promising tool for future research. It could be used to monitor signal amplitude, frequency content, and muscle recruitment patterns during stimulation, helping to fine-tune FES protocols to reduce fatigue and improve long-term outcomes.

2. *"Volitional EMG Controlled Wearable FES System for Lower Limb Rehabilitation", J. Jung, D.W. Lee, Y. Son, B. Kim, J. Gu, H.C. Shin , 2021* [3]

   This study presents a wearable Functional Electrical Stimulation (FES) system specifically designed for lower limb rehabilitation, with a focus on enhancing user responsiveness and natural movement. Unlike traditional FES systems that operate on fixed or cyclic stimulation patterns, this approach centers on real-time modulation of stimulation based on the user's voluntary movement intention, detected through electromyographic (EMG) signals.
   The device is lightweight (1.5 kg including battery), portable, and user-friendly. Its key innovation lies in the integration of EMG signals to continuously adapt the stimulation intensity, creating a closed-loop system that aligns more closely with the user's motor intent. The system includes:
   - EMG sensors placed on lower limb muscles to capture voluntary activation signals.
   - Inertial Measurement Units (IMUs) for tracking joint angles, movement dynamics, and body orientation.
   - Micro-controllers and communication units to process data and manage stimulation control.

   The control architecture involves three main steps:
   - Volitional EMG Estimation: A novel signal processing method isolates voluntary EMG (vEMG) from signals contaminated by stimulation artefacts, allowing the system to reliably detect intentional

muscle activity even during ongoing FES.
  - Intention Detection: The user's motor intent is quantified through the RMS value of the vEMG. When this value crosses a defined threshold, the system interprets it as an intention to move and adjusts stimulation accordingly.
  - Stimulation Modulation: The FES current is modulated in real time in proportion to the vEMG RMS, supporting exercises like squats, gait, or heel raises with stimulation that matches the user's voluntary effort.

This volitional control paradigm enables more natural, functional movements during rehabilitation, supporting patient engagement and safety while enhancing the efficacy of muscle activation.

3. *"Recognition of Human Lower Limb Motion and Muscle Fatigue Status Using a Wearable FES-sEMG System", W. Zhang, Z. Bai, P. Yan, H. Liu and L. Shao , 2024* [4]

   This paper presents an advanced wearable FES system aimed at rehabilitation, motor training, and assistive support for lower limbs. Its key innovation lies in combining textile-based sEMG sensors with a custom deep learning model (FES-sEMGNet) to enable real-time recognition of both user motion intent and muscle fatigue. These insights are used to adapt the electrical stimulation parameters dynamically, ensuring more effective and safer interventions.

   A major contribution is the development of 3D-knitted textile electrodes, designed to be flexible, breathable, and suitable for prolonged use (overcoming the discomfort and short lifespan of traditional gel electrodes). Integrated with these textile elements are surface EMG sensors that capture muscle activity even during electrical stimulation, enabling continuous monitoring.

   The intelligent control framework relies on FES-sEMGNet, a deep neural network optimized for real-time processing on embedded hardware. It consists of two parallel branches: one dedicated to classifying lower limb movements (e.g., walking, flexion, extension), and the other to assessing muscle fatigue levels (from relaxed to severely fatigued). Rather than relying on handcrafted features, the network processes time–frequency sEMG images to autonomously extract relevant patterns for both motor and fatigue recognition.

   The system's control strategy integrates these dual outputs to adapt stimulation in real time:
   - Motion detection ensures that FES intensity, timing, and pattern match the intended movement.
   - Fatigue monitoring enables responsive changes such as reducing stimulation or pausing activity to prevent overuse injuries.
   - The architecture supports autonomous, on-device decision-making with low latency, allowing the system to respond instantly and reliably to changing physiological states.

   Overall, this work demonstrates a significant step toward personalized and intelligent FES systems, where wearable comfort, muscle-state awareness, and adaptive control are tightly integrated to enhance rehabilitation outcomes.

4. *"Garments for functional electrical stimulation: Design and proofs of concept", B. Moineau , C. Marquez-Chin , M. Alizadeh-Meghrazi, M. R Popovic, 2019* [5]

   This study explores the development and proof-of-concept validation of wearable garments (specifically a shirt and a pair of pants) integrating knitted electrodes made from conductive yarns, aimed at delivering Functional Electrical Stimulation (FES) for individuals with paralysis. The research addresses common limitations of conventional gel electrodes, such as discomfort, limited re-usability, and the need for manual positioning.

   Using seamless knitting technology, the garments were constructed from elastic materials (nylon and

Lycra), embedding silver/silver-chloride-coated nylon yarns to form conductive zones directly into the fabric. The electrode layout was informed by anatomical landmarks, and the garments were designed to allow consistent positioning and pressure without external taping.

A variety of electrical, functional, and usability tests were carried out, primarily using an able-bodied volunteer to assess performance. Fabric electrodes were compared to traditional gel electrodes under different conditions, particularly focusing on stimulation thresholds, comfort, and practicality in daily use.
Key findings include:

- Electrical performance: Fabric electrodes were effective at delivering muscle contractions, especially when freshly moistened. However, their performance declined rapidly as they dried, becoming uncomfortable after 14 minutes post-moistening.
- Comfort and usability: Garments offered improved electrode stability during movement, and easier donning and repositioning, making them particularly promising for users with limited mobility or dexterity.
- Limitations: The primary limitation of fabric electrodes was the dependency on moisture for effective conductivity, along with variation in stimulation thresholds due to inconsistent hydration levels in the fabric or body.

Overall, the study demonstrates the feasibility and advantages of wearable FES garments with integrated textile electrodes, but also underlines the need for future design improvements to overcome hydration and longevity challenges.

Parameter	**Fabric Electrodes (Knitted)**	**Gel Electrodes (Traditional)**
Initial Comfort (wet)	High (comparable to gel electrodes)	High
Comfort over time	Decreases after 9-14 minutes (drying)	Stable
Stimulation Voltage	Significantly rises as electrodes dry	Stable
Sensory Threshold	Lower (subjects more sensitive to stimulation with textile electrodes)	Higher
Max Stimulation Threshold	Higher (less tolerance at high intensity)	Lower
Stability during movement	High (due to garment elasticity)	Lower (may peel or shift)
Ease of Positioning	Easy (can be worn independently)	Low (requires manual positioning, taping and expertise)
Ease of Donning	Easy (can be worn independently)	Complex (requires manual setup)
Moisture Dependency	High (requires water for conductivity)	Medium (requires pre-gelling before use)
Lead Management	Integrated and secure	External and prone to disconnection

Table 1: *summary of electrodes performance & usability tests.*

5. *"A review of textile-based electrodes developed for electrostimulation", L. Euler , L. Guo, N.K. Persson, 2021* [6]

   Euler et al. (2021) offer a comprehensive review of textile-based electrodes developed for electrostimulation, focusing on construction techniques, design strategies, and the evaluation of pain and discomfort during use. The study highlights the growing interest in integrating electrodes into garments for rehabilitation, with an emphasis on enhancing wearability and user compliance.

   Regarding construction methods, the review outlines various conductive materials such as silver-coated fibers, carbon-based compounds, and conductive polymers. These materials are chosen not only for their electrical properties but also for their comfort and compatibility with the skin. Multiple fabrication techniques are described (knitting, weaving, embroidery, and conductive coatings) each allowing different degrees of flexibility, breathability, and integration into garments. Flat, flexible electrode geometries are

emphasized to enhance skin contact and reduce discomfort.

In terms of design proposals, the paper underlines the importance of seamless integration into garments, aiming to improve usability, aesthetics, and patient compliance. Both wet and dry electrode systems are discussed: wet electrodes provide high conductivity but require conductive gels, whereas dry electrodes offer improved convenience but currently face limitations in long-term performance.

The review also delves into the evaluation of comfort and pain, recognizing it as critical for effective therapeutic use. Quantitative methods (i.e., questionnaires) and qualitative tools (i.e., interviews) are proposed to assess user experience. Factors such as electrode stickiness, moisture retention, and surface irritation are identified as potential sources of discomfort. The authors advocate for standardized comfort metrics and call for more research on the correlation between electrode design features and user-reported pain.

In conclusion, the review identifies promising directions for making textile-based electrical stimulation more wearable, effective, and acceptable in clinical or home-based rehabilitation settings.

All the articles presented were used to understand the baseline of research related to FES and EMG smart garments. Insights for the developed concept design were extracted regarding the properties of textile electrodes, the muscular activity detection and the control strategies to implement both EMG monitoring and functional electrical stimulation in a single garment.

2.3.2. Patents

The analysis of the available patents was conducted to have a complete view of the research activity related to electrical stimulation and electromyography embedded into smart garments. Tools such as Google Patents and Espacenet were used for the searching activity.
Several patents were found to include both electrical stimulation and EMG monitoring: patent US11,524,160 B2 [7] proposed an arm sleeve that embeds the control of functional electrical stimulation through the analysis of the motor unit action potential recorded by EMG electrodes, letting synergistic use of the electrical monitoring and stimulation, while patent US2023/0277109 A1 [8] proposed a similar solution but with the difference that the two activities are used separately, without the EMG monitoring used as control.
Patent US 2019/0151649 A1 [9] developed the concept of a leg orthosis to be placed just below the knee joint. The device is thought to stimulate the muscles during walking for a correct gait execution in people with physical impairments, and this is possible through the connection of the orthosis with a pressure sensor placed on the insole of the shoes.
Eventually, examples more inherent to the project topic were found: patent WO2025/064963 A1 [10] showed a smart suit with embedded electrodes that perform both electrical stimulation and monitoring thanks to the use of a switch system, avoiding the saturation of the EMG system. A similar concept was proposed in the shorts of the patent CN 221771318 U [11], in which external electrodes were attached to the garment through a control unit placed at the level of the pelvis. The most complete idea was presented in the patent US 2024/0238584 A1 [12] where the positioning of the electrodes, of the leads and of the connections were displayed on smart pants for a complete stimulation of the muscles present in the lower limbs. Moreover, the patent showed the correct alternation of electrical stimulation of each muscle for an effective induced gait execution.
All the patents just presented were used to understand the baseline of research related to FES and EMG smart garments for a possible future positioning of the project's product as alternative or competitor, and also to consider new insights such as the positioning of the electrodes or the type of available controls.

2.3.3. Market Analysis

The last section of the research phase deals with the market analysis that was performed to look at the current available smart garments always related to electrical stimulation and electromyography. Shorts, pants or suits were the primary subjects of the research to find possible insights for the project purpose. The exploration was divided between the two functionalities because of the absence of products that offer both of them; the only dual device that was found is Stella Bio [13] from EGZO Tech (Figure 3), a control unit worn with a belt that is attached to surface dry electrodes through external cables. The device is mainly used for rehabilitation to reduce spasticity, improve coordination and regain lost functions, however lacking of the possibility to be worn quickly and with ease.

Regarding EMG smart garments, four products were found in the analysis and their specifications can be observed in Figure 2. All the products appeared to be really similar in the main properties: shorts with an average of six or eight dry textile electrodes (depending on the number of monitored muscles) that connect to detachable control units through internal cables embedded into the garment; moreover, the material for the shorts was found to be elastic tissue for the correct positioning and stabilization of the electrodes. No further details about it were explicitly shared ([14],[15],[16],[17]).

On the other hand, a variety of different products were found for the electrical stimulation functionality (Figure 3). Suits like Teslasuit [18], Exopulse Mollii Suit by Ottobock [19] and Antelope Evolution EMS [20] are products thought for a total body stimulation which can include EMS, TENS or FES. The electrodes used for these devices are mainly dry and their number ranges from few ones (six or eight for Antelope Evolution EMS) up to a large quantity of 58 for the Ottobock product and 114 for the Teslasuit, while the leads are both internal or external and the control units are detachable or wireless connected, like in the Teslasuit case. Concerning shorts for electrical stimulation, Berkel Bike Shorts [21] and BioShortX by AxioBionics [22] were observed on the market. The first ones are used for FES and embed gel electrodes that are connected to a pocket-sized control unit through embedded cables. The second ones instead use Biogel or Ag-based electrodes for TENS, NMES and FES; the connections are internal and the control unit is detachable. For both suits and shorts the main material used for the garment were found to be polyamide (nylon) and elastane (spandex or Lycra), because of their elastic and compressive properties. Eventually a different device was also found to be used for FES: Ottobock L3000 [23], a leg orthosis used for gait execution and synchronized with a pressure sensor placed in the insole of shoes. It is characterized by hydrogel electrodes, wireless control unit and polymeric material for the textile part of the orthosis.

From the table of Figure 3 it is also possible to notice that the range of parameters used for the stimulation (frequency, voltage, current, pulse width) varies depending on the specific application.

The market analysis turned out to be effective to understand the main characteristics of the present products and to retrieve insights, such as the material of the garment (nylon and elastane), the type of electrodes (dry textile) and the connections (embedded leads with detachable control units).

	Myontec Mbody 3	Athos	ITRI	STRIVE sense3
type	EMG	EMG	EMG	EMG
type of electrodes	dry textile	dry textile	non-skin contact	dry textile
n° electrodes	6	8	-	8
n° channels	6	-	4	-
frequency	x	x	x	x
voltage	x	x	x	x
current	x	x	x	x
pulse width	x	x	x	x
control unit	detachable	detachable	detachable	-
connections	internal cables	seamless	-	embroided
material	elastic tissue	elastic tissue	elastic tissue	elastic tissue
sizes	5	6	-	-

Figure 2: *summary table of EMG devices on market.*

	Berkel Bike Shorts	Teslasuit	Stella Bio	Ottobock L300	BioShortX by Axio Bionics	Exopulse Mollii Suit by Ottobock	Antelope Evolution EMS
type	FES	EMS, TENS,FES	EMG + EMS,TENS,FES	FES	TENS,NMES,FES	TENS	EMS
type of electrodes	gel	dry textile	dry	hydrogel; viscose+polypropilene+ stainless steel	Biogel + silver	silicone rubber	Dry silicone electrodes
n° electrodes	6	114 (52 trousers + 62 jacket)	variable	2 (upper leg) + 2/3 (lower leg)	6	58	6/8
n° channels	-	80 (32 trousers + 48 jacket)	4+2	-	2/4	40	-
frequency	-	1-150 Hz	-	10-45 Hz	25 Hz	20 Hz	1-150 Hz
voltage	-	1-60 V	-	1-130 V	-	-	1-120 V
current	-	1-150 mA	-	0-100 mA	-	-	1-240 mA
pulse width	-	2-100 us	-	2-900 us	300 us	25-175 us	5-500 us
control unit	pocket	wireless control from application	belt	control unit (wireless) + EPG (detachable)	detachable	detachable	magnetic holder
connections	internal cables	internal cables	external cables	internal cables	internal cables	external cables + magnetic coupling	internal cables
material	elastic tissue	polyester/elastan/nylon/ spandex	-	tissue/polymer	elastic tissue	polyamide/elastane blend	polyamide/ polypropylene/ elastane
sizes	-	6	-	6	personalized	37	5

Figure 3: *summary table of electrical stimulation devices on market.*

2.4. Refined Objective

The research phase helped to refine and consolidate the objective of the project giving new insights and hints for the creation of a product different from the already available ones. The previous step strengthened the hypothesis to develop a garment for both electrical stimulation and EMG monitoring, because of the lack of similar devices on the market, while only few examples in the patents and in the literature.
Additionally, the implementation of inertial measurement units (IMUs) into the control devices is included in the final objective to guarantee a simultaneous use of the electrical stimulation system and the control units. The addition of inertial sensors could supply the monitoring activity with further information. This approach can be also found in the Teslasuit product [18], used for both stimulation and movement tracking.
Eventually the project's objective remain similar to the previous presented, with the consolidation of the purposes and the addition of further elements, like the IMUs.

2.5. Requirements

The last phase of the brief formulation relied on the drawing up of the requirements, necessary to properly satisfy all the needs and key points that define the final outcome of the project. The list was also based on the findings coming from the research activity. The requirements were divided into functional, non-functional and optional depending on their relevance. In the following tables all the three types of requirements are displayed, with possible preliminary solutions and specifications. Some of the details related to the FES electrodes were directly taken from the team involved in the project Active^3 – Everyone,Everywhere,Everyday.

FUNCTIONAL REQUIREMENTS:

Description	**Method-Specifications**
Shape and size of FES electrodes must be large enough to stimulate the correct muscular compartment and avoid pain in the user due to current concentration.	Rectangular shape (5x9 cm or 4x8 cm) with rounded corners to avoid current concentration. See subsection 3.1.
Impedance of FES electrodes must be low to guarantee the correct transmission of the stimuli.	Maximum skin-electrode impedance (20-30 kΩ for wet, <200 kΩ for dry)[24],[25].
Placement of FES electrodes must correctly stimulate the chosen muscular compartments depending on the different solutions (gluteus maximus, rectus femoris, biceps femoris, gastrocnemius medialis).	Electrodes must be placed on the innervation zones of the muscles (distal and proximal ends). See subsection 3.1.
The connector between the pants and the stimulator leads must be placed in a comfortable position, also quick to be reached.	Lateral part of the thigh and integrated with the textile (snap button solution).
Placement of the EMG electrodes must correctly acquire the desired muscular electrical activity (gluteus maximus, rectus femoris, biceps femoris, gastrocnemius medialis).	SENIAM protocol. See subsection 3.1.
Size of the EMG electrodes does not have to be too large to monitor the only desired muscles. Moreover, the direction of the electrodes must be longitudinal to the muscle's fibers.	Recommended round shape with diameter between 10 and 20 mm [4]

Description	Method-Specifications
Impedance of EMG electrodes should be low and should be similar between two electrodes used in bipolar configuration to guarantee proper acquisition of the signal.	Maximum skin-electrode impedance (20-30 kΩ for wet, <200 kΩ for dry)[24],[25].
Pre-amplification and filtering of the EMG signals must increase the SNR to have a good acquisition of the activity.	Gain: from 0-5 μV to 0-5 V. CMRR: 90-130 dB Band-pass filter (20-500 Hz). Notch filter (50/60 Hz)[26],[24].
The EMG control units must be able to process and transfer data acquired from all the EMG sensors with the correct accuracy. The same should be guaranteed by the IMUs.	SRAM/Flash memory. Low energy consumption. BLE (>= 1Mbps). Sampling rate >= 1000 Hz. ADC resolution >= 12-16 bit. IMU sampling rate >= 500 Hz.
The battery of the control unit must guarantee the functioning of each component for an ideal life of activity of 2-3h.	Rechargeable Li-ion Polymer Battery (3.7 V, 500 mAh).
The material of the garment must guarantee tensile elasticity to fit the several users and compressive one to ensure the positioning of the electrodes.	Polyester, cotton, elastane (spandex, Lycra), polyamide (nylon), polypropylene.
The user interaction with the system, in terms of correct placement/connection of the electrodes and control units, must avoid misunderstandings and be quick and effective.	Use of different color markers and easy-to-read indicators for position and direction.

Table 2: *functional requirements.*

NON-FUNCTIONAL REQUIREMENTS:

Description	Method-Specifications
The FES electrodes configuration must guarantee precise muscle selectivity.	Bipolar configuration.
The EMG electrodes configuration must ensure the correct muscle monitoring avoiding crosstalk.	Bipolar configuration and IED (inter-electrode distance) = 20 mm. [4]
Possibility to reduce impedance of the FES electrodes to avoid localized pain/burning.	Use of electrolyte between skin and electrodes (water, hand cream).
The product must propose different sizes to fit most of the users.	Three main categories of sizes (S, M-L, XL-XXL).
The garment must be washable, still guaranteeing correct functionalities.	Waterproof textile and protection of conductive connections.
The garment must guarantee proper thermal and sweating regulation.	Presence of breathable textile parts and sweating analysis of the body segments.

Table 3: *non-functional requirements.*

OPTIONAL REQUIREMENTS:

Description	Method
The monitoring of the fatigue should be done offline from the data retrieved by the control unit.	Use of coding applications (Matlab, Python languages...).

Table 4: *optional requirements.*

3. Ideation

3.1. Electrodes Positioning

This first section of the ideation phase regards the identification of the correct positioning of the electrodes, both for functional electrical stimulation and electromyography. The process that led to this result was structured into a series of successive phases. Specifically, the most important steps involved consisted in consulting the SENIAM and BMC's studies guidelines [27] [28], identifying the electrodes shapes and dimensions, and, lastly, asking to some of the researchers at Politecnico di Milano to seek guidance on best practices to be adopted.

The correct placement of electrodes, for both FES and EMG monitoring, is crucial: improper positioning can significantly compromise signal quality or, in the case of stimulation, raise safety concerns. To mitigate such risks, it is essential to rely on established and reliable guidelines.

EMG Electrodes
Regarding EMG electrodes, their correct placement was determined based on the guidelines provided by the SENIAM project. Indeed, SENIAM, a European research initiative focused on surface electromyography, offers standardized and widely accepted recommendations for the accurate positioning of EMG electrodes.
Based on the identification of specific anatomical landmarks and the selection of motor points corresponding to the innervation zones of the target muscles (gluteus maximus, rectus femoris, biceps femoris, gastrocnemius medialis), along with knowledge of the orientation of muscle fibers within the relevant muscular group, it was possible to determine the theoretical optimal placement of the muscle bellies. This process was guided by established recommendations (as detailed in [27]) related to the distance of the muscle's belly from the reference anatomical landmark or the innervation point of each muscle. These locations represent the most appropriate sites for detecting signals related to muscular activity.

After identifying the optimal sensor placement, a detailed definition of their orientation, shape and dimension, and relative positioning was carried out. A bipolar configuration was adopted for each EMG sensor: an anode and a cathode electrode were placed along the longitudinal axis of the targeted muscle, maintaining an inter-electrode distance (IDE) of 20 mm [4]. Regarding the electrodes shape and dimension, findings from several studies were taken into account: circular electrodes (with a ∅10 mm diameter) have demonstrated superior accuracy and stability in monitoring muscle activity.
Figure 4 illustrates an example of the visual guidelines provided by the SENIAM project for the targeted muscles [4] [27].

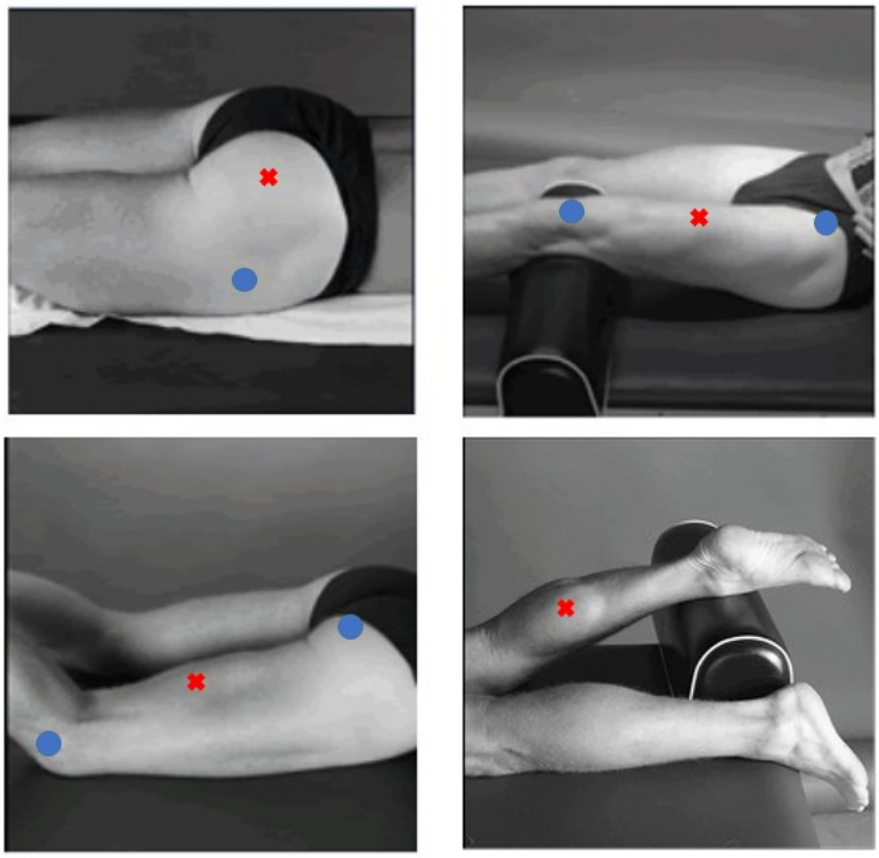

Figure 4: *positioning scheme for EMG electrodes (SENIAM – gluteus maximus, rectus femoris, biceps femoris, gastrocnemius medialis). The blue points are the anatomical landmarks considered, while the red crosses represents the targeted muscles' bellies.*

FES Electrodes

Alternatively, regarding the actuators for functional electrical stimulation, their correct positioning was determined based on dual sources: the recommendations and information of the PhD researchers actively working on projects involving trike and FES, together with the guidelines provided by some studies published on BMC platform [28]. As a matter of fact, by exploiting the expertise of the members actively involved in research projects on the topic of functional electrical stimulation, and by evaluating the state of the art on BMC open-access platform, it was possible to identify the correct positioning of the electrodes to realize an effective stimulation of target muscles (gluteus maximus, rectus femoris, biceps femoris, gastrocnemius medialis).

A bipolar configuration is also required for electrodes used in Functional Electrical Stimulation (FES). In fact, effective stimulation relies on a pair of electrodes (an anode and a cathode) between which an electric current is generated for each stimulation pulse. The current flows from the anode to the cathode, along the dipole created by the two electrodes.

In the context of FES, both the geometry and dimensions of the electrodes, as well as the chosen control strategy, play a crucial role. Based on findings from scientific literature [2] [29], rectangular electrodes were selected, with a standardized size of approximately 4 cm in height and 8 cm in width. This configuration balances the need for efficient muscle fiber recruitment with user comfort and usability. Additionally, rounded corners were preferred, as they help deliver higher voltages while maintaining acceptable levels of comfort and minimizing the perception of pain during stimulation [2].

A detailed analysis of the stimulation control strategy was also carried out, as different control modes imply changes in hardware and sensor requirements. Given the objective to integrate this system with technologies already developed within the Active[3] project, a bipolar external control strategy was selected. The stimulation

pulses are generated using the stimulation hardware already in use in the project, allowing also full leverage of both the theoretical and technical expertise of the research team.

The discussion with the trike project team was also essential in addressing potential electromagnetic interference between the FES and EMG systems. As a result, the two electrode typologies were designed to operate independently and remain spatially distinct. The FES electrodes were positioned in a dedicated configuration, defined in close collaboration with the PhD researchers at Politecnico di Milano. This setup not only maximizes the amount of muscle tissue activated (thanks to the relatively large inter-electrode distance), but also avoids overlap with the more sensitive EMG electrode placement zones, thereby ensuring optimal performance for both systems.
Figure 5 illustrates the adopted electrodes placement map for functional stimulation.

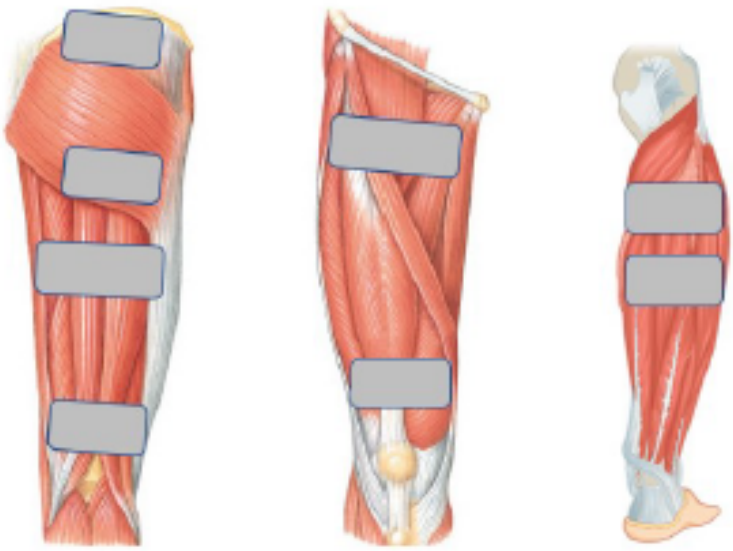

Figure 5: *positioning scheme for FES electrodes. From left to right, respectively, are presented the electrodes positioning for gluteus maximus and biceps femoris, rectus femoris, and gastrocnemius medialis.*

3.2. Sewing Pattern

After identifying the theoretical positions for both the EMG and FES electrodes, a manual process was conducted to determine their practical placement. This was performed using Decathlon leggings composed of 92% cotton and 8% elastane, which served as the base for marking all electrode positions.

This activity had two main objectives: to obtain a comprehensive overview of the component layout for both EMG and FES systems and to verify the absence of overlaps, as well as to collect fundamental data for evaluating the scalability of the sensor positioning across different garment sizes. Furthermore, it provided a hands-on opportunity to engage with the physical product and enhance design familiarity.

Following the SENIAM guidelines and the previously defined criteria, the main anatomical landmarks were identified via palpation on a subject wearing the leggings. These landmarks enabled the precise localization of both the EMG and FES electrodes. Adhesive tape was used to mark the positions, which allowed for a visual inspection of spacing and potential interferences, taking into account the hardware footprint of each component. Figure 6 shows the result of the practical mapping of electrode positions carried out on the available leggings.

Figure 6: *results of the practical mapping of EMG and FES electrodes.*

Subsequently, using a tailor's measuring tape, the distances between electrodes and key garment references (such as the waistline and the vertical crotch axis) were measured. The overall dimensions of the leggings were also recorded and used to reproduce a digital garment pattern via Autodesk Inventor. The resulting CAD model is presented in Figure 7.

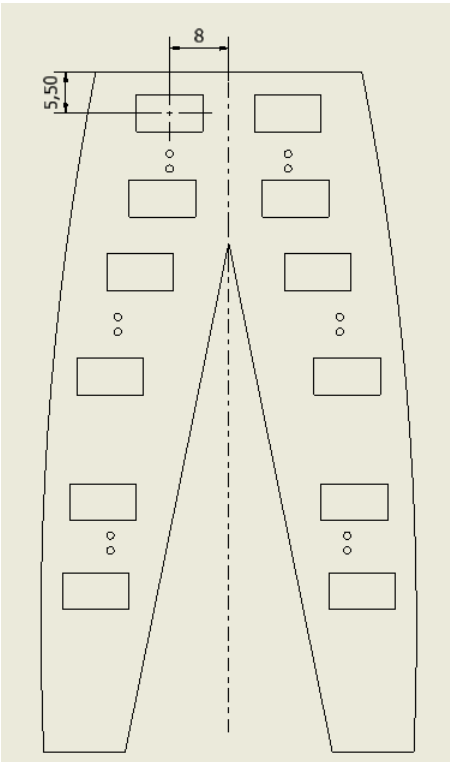

Figure 7: *digital CAD sewing pattern - Back View. Example of how the positions of the electrodes were extracted.*

The decision to develop a digital pattern was driven by both practical considerations and the need to study the scalability of the electrode layout across multiple garment sizes. Since the intended application targets a broad user base, accounting for different body morphologies was essential. The digital format allows for flexible adaptation of the electrode placement according to size variations.

A market analysis (conducted both online and in-store at Decathlon) allowed to collect dimensional data for the various available sizes of unisex sports leggings, which are reported in Table 5. Comparing these with the size L leggings used during testing, it was found that the most significant differences occur in the waist circumference, while leg length (outseam) remains relatively stable.

Dimension/Size	**S**	**M**	**L**	**XL**	**XXL**
Outseam	89cm	91cm	92cm	93cm	95cm
Ankle Opening	10cm	11cm	11cm	12cm	12cm
Front Rise	21cm	23cm	23cm	23cm	24cm
Waist Circumference	90-96cm	97-102cm	103-108cm	109-116cm	117-124cm

Table 5: *Decathlon pants' dimensions.*

This finding led to the conclusion that the electrode configuration validated on size L could reasonably be extended to other sizes, especially considering that the most critical spacing is along the direction of muscle fibers, which corresponds to the leg length (that varies minimally). In addition, due to the high elasticity of the technical fabrics used for such garments, minor positional deviations are unlikely to compromise system performance and were thus deemed acceptable in the design.

Nevertheless, a clarification is necessary. While a slight misalignment in FES electrodes may only lead to a modest reduction in stimulation efficiency (i.e., recruitment of fewer muscle fibers), EMG electrodes typically require more precise localization. However, due to the current lack of detailed anthropometric data on elderly subjects (the project's primary target group) and the absence of automated tools for identifying optimal EMG electrodes positions on elastic garments, it was concluded that a small accuracy loss could be tolerated. This is consistent with the product's intended use, which does not require high-fidelity performance tracking.
To scale the positions on different sizes, an idea could be to define the coordinates of each electrode as a percentage of distance from the top (i.e. waist-line) with respect to the total height of the pants (in the vertical direction) and as a percentage of distance from the vertical symmetry line with respect to the half-width of the pants (in the horizontal direction). However, this solution is quite approximate and other standardized methods should be developed and used.

In conclusion, the L-size digital model will be used as the design reference for the subsequent development phases.

3.3. Materials & Processes

In conjunction with the other steps of the ideation phase, the reasoning about the choice of the materials to use for the design concept was a key point.
The first thought concerned the selection of the production process and materials for the pants realization. The two main technologies adopted for the production of sportswear are circular knitting and warp knitting seamless that bring with them important advantages with respect to the common cutting and sewing processes: decrease of the waste, faster production, reduction of stitching and so more comfort, and eventually high level of fitting and elasticity.
By a comparison between the two seamless options, the warp knitting turned out to be the most appropriate solution for the development of the pants. This technology permits to avoid completely the presence of stitching due to the possibility to create at the same time the two tubular parts of the pants and to connect them directly,

while for the circular knitting this is not possible, requiring a further step of cutting and sewing of the inner part of the garment. Moreover, the warp knitting technique leads to better resistance to damages due to the different disposition of the yarns and subsequently to the creation of complex and breathable designs. As last advantage, the machines used for warp knitting allow to produce more than one garment at a time, reducing the production time despite requiring a higher initial investment.
Concerning the materials, a combination of polyamide (nylon) and Lycra (elastane) were decided to be the most suitable. The choice was based on the multiple results obtained by the previous market analysis (see section 2.3.3), by an online research of the most used fibers for sportswear and by looking at present articles related to the prototyping of smart garments. Examples can be found in the works of Paiva et al. [30],[31] or Ferri et al. [32] where composition of polyamide and elastane was adopted.
The combination of these fibers allows at the same time to have good elasticity of the garment for fitting different shapes and sizes of the body, and to guarantee high compressive properties which are required for the correct contact between the skin and the electrodes. Additionally, they present the required thermal resistance for the thermoadhesion of the electrodes and tracks to the pants (up to 180°C for nylon and 195°C for elastane).

The second part concerned the selection of the type of electrodes. A sequential reasoning was performed before arriving to the appropriate solution. The initial idea consisted in the use of Elastatrode ink-based textile electrodes proposed in the previously presented Master's thesis of Valentina Olivieri [2] because already tested by the team working on the trike and the FES, bringing to no statistical differences with hydrogel ones. However, the following study of Guzzardi [33], related to the mechanical resistance and washability of the same electrodes, highlighted some limitations like the fragility of the tracks and the increase of resistance after repeated cycles of washing test. For the purposes and requirements of the actual project a good durability to stretching and to washability are fundamental, therefore the first idea was discarded.
Subsequently the hypothesis of implementing textile electrodes and their tracks through embroidery of conductive yarns or by coupling different textile layers to add on the garment was taken into consideration. Nevertheless a significant limitation was faced in this case: the drastic reduction of breathability and elasticity of the pants. The cause is the necessity to cover and protect with wide patches of thermo-adhesive layer all the electrodes' tracks that in the final design would occupy large part of the garment. This could result into enhanced discomfort for the user and mainly into reduction of breathability which is essential during physical activity.
To overcome the problems of durability (mechanical and of washability) and of breathability a suitable solution was eventually implemented.
Nanoleq technology [34] offers three different components for the integration of electrodes into smart garments: ElectroSkin GECKO electrodes, PhantomLink and PhantomTape X (Figure 8).
ElectroSkin GECKO are textile electrodes composed by a thermally adhesive layer to glue them to the garment, an internal conductive layer (materials were not specified for privacy; skin-electrode impedance < 150 kΩ) to transmit electrical signals and an external layer of silicon rubber to enhance the adhesion with the skin (Figure 9). The advantage of this product is the possibility to be used both for FES and EMG applications, and to be personalized on request. Moreover, they are tested for cytotoxicity (ISO 10993-5:2009), skin sensitization (ISO 10993-10:2021) and irritation (ISO 10993-23:2021), and they guarantee reliable functioning even after 100 cycles of washing test.
PhantomLink is the element that allow safe connection between the electrodes and the tracks (PhantomTape X) or the snap buttons for external linking. It is composed by a thermo-adhesive layer, a conductive one and an insulating one, for a total thickness of 0.38 mm (Figure 10). The materials used are polyamide, silver and polyurethane. Also in this case it is tested to resist 100 cycles of washing test and to accept length and width's elongation of 60% and 5% respectively.
PhantomTape X is the cable that let electrodes to be connected to far zones. As previously explained, it can be linked to electrodes or external connections through PanthomLink. It is composed primarily by a stretchable helical conductor (two strands of Ag-plated Cu micro wrapped around synthetic textile filaments) with an elongation up to 80%, closed between a thermo-adhesive layer and insulating textile (Figure 11). The total width of the element is 10 mm, while the thickness is 1.5 mm. The washability is guaranteed and it was also tested by Rotzler [35] in his analysis of washing behavior of textile integrated circuits. The results showed the durability

to 100 cycles of washing test with the only increase of resistance due to damages in the connection with snap fasteners (to notice the lack of the use of PhantomLink).
All of the three components can be integrated onto the pants by lamination at 130-150°C and 50 – 100 g/cm^3 for 20-30 seconds, guaranteeing the thermal resistance of the fiber chosen for the pants.
By selecting this solution, the requirements of electrical (skin-electrode impedance < 150 kΩ), mechanical and washing durability, and of breathability (reduction of area occupied by electrodes and tracks) can be satisfied.

Figure 8: *composition scheme of Nanoleq technology.*

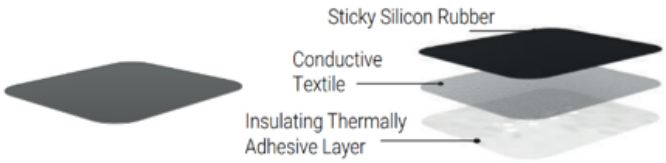

Figure 9: *ElectroSkin GECKO composition.*

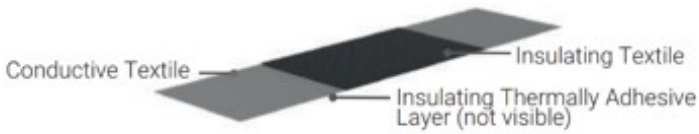

Figure 10: *PhantomLink composition.*

Figure 11: *PhantomTape X composition.*

3.4. Leads & Control Units

With the aim of designing a pair of pants equipped with systems for surface electromyographic (EMG) monitoring and functional electrical stimulation (FES), intended for autonomous use, it was immediately clear that a dedicated solution was required for the integration of the conductive traces connecting the electrodes to their control systems. The main objective was to minimize the presence of external wiring, which could interfere with the user's motion and pedaling pattern, and increase the risk of sensor disconnection and system malfunction.

A dedicated study was carried out on the optimal placement of the traces presented in the previous section (PhantomTape X) within the garment. In order to maintain optimal electrode positioning (for both EMG and FES), minimize interference with movement, and avoid placing any wiring or adhesives on the inner thigh area (which is prone to higher skin sensitivity and sweating), a functional and rational design layout was developed. As shown in Figure 12, the electrodes and conductive paths were grouped by muscle region (glute/pelvis, thigh, tibia/calf), achieving three key goals: simplifying connector interfaces, enabling independent evaluation of each segment during cycling, and optimizing the trace routing (shorter and simpler paths).

Figure 12: *scheme of conductive paths (left) and outer snap buttons placement for external connections (right).*

The choice to route the traces to the outer thigh area for their connection to external devices served three purposes: simplifying the connection process, minimizing motion interference, and exploiting the lower skin sensitivity in that region (stimulation might otherwise cause irritation in other areas).

Regarding the FES control system, it was decided (following discussions with the research team at Politecnico di Milano) to retain the stimulation controller already integrated into the trike. This system, widely validated, ensures synchronization of each muscle's stimulation with the pedaling pattern via real-time crank angle tracking. The FES electrodes' conductive traces were thus designed to terminate in individual snap-fit connectors, enabling fast connection to the external wiring (Figure 13).

Figure 13: *FES external wiring.*

A different approach was taken for the EMG monitoring system. In addition to capturing muscle activation signals, it was essential to analyze pedaling quality (whether self-initiated or FES-induced) by tracking the time evolution of joint angles across the three body segments through the use of IMUs. A market survey identified OT Bioelettronica Quattro+ [36] as a promising control unit featuring both EMG recording and embedded IMUs. However, its reliance on external cabling for electrode connection posed a limitation.
To address this, the design of custom control units was proposed. These would integrate both IMUs and snap-fit connectors for direct attachment to the conductive traces of the EMG electrodes. A total of six units were foreseen, each capable of managing 2 (glute/calf) or 4 (thigh) connections, avoiding the need of an additional linking to a reference electrode thanks to a differential acquisition given by the bipolar configuration. The devices should include all the electronic components to fulfill the requirements presented in subsection 2.5, specifically a pre-amplifier, an ADC unit, a microcontroller with memory, IMU, BLE unit in case of connection with a dedicated application, and a rechargeable battery. The control units also present a USB-C port for recharge and an external button for switch on/off and control activities, with a LED. The external case measures 50x30x10 mm (the size could be changed depending on the chosen specific electronic elements). A future development of the current project could deepen the analysis and the proper design of the control units, proposing and prototyping specific components and their connectivity. Figure 14 shows the prototype developed.

Figure 14: *prototype of EMG control units. Details of snap connections and interactive elements.*

3.5. User Interfaces

The development of a wearable device designed to be used autonomously by the identified target users requires not only a strategic and carefully studied placement of control units and electrode connections, but also a thorough analysis of the user interfaces (UI). Ensuring that elderly users can operate the system independently and intuitively is essential, with the goal of minimizing errors related to control unit placement (for EMG monitoring) and cable connection (for functional electrical stimulation, FES).

To identify the most effective UI strategies to be integrated into the product, a literature review was conducted focusing on interface design for elderly users. In particular, the study by Zhuo et al. ([37]) provided valuable insights into effective UI design principles for this demographic.

The article highlights the effectiveness of simple, linear interfaces that reduce distractions and facilitate faster task completion. It also emphasizes the importance of breaking down complex tasks into simpler, more manageable steps. The use of icons is shown to be highly effective: when appropriately sized and positioned, they enhance intuitive understanding of key functions. Furthermore, the study demonstrates that both the color scheme and the spatial arrangement of icons play a crucial role in improving interface usability. A well-organized layout with similarly shaped icons, differentiated by color and set against a high-contrast background, supports focus, attention, and overall UI clarity.

Based on these findings, the decision to group EMG and FES electrode connectors by muscle group (corresponding to the control unit attachment points) was made to streamline the connection between sensors and the control system. Regarding the control units themselves, a minimalist yet user-friendly design was developed: indicators clearly specify the correct height, orientation and side of the leg for each unit, making the system intuitive and accessible. A visual representation of a control unit is shown in Figure 15.

Figure 15: *: EMG control units. Details of user interfaces.*

Similarly, the connection points for the FES system were designed with particular attention to the needs of elderly and non-expert users. For each bipolar electrode pair (used to stimulate individual muscles), a color-coded system enables users to connect the cables correctly on both legs. The choice of the colors was made to maintain a correspondence with the already used ones in the stimulator's cables, as previously shown in Figure 13. Additionally, placing these connection sites adjacent to the EMG control units helps to simplify product usage. Regarding the user's detection of positive and negative electrodes for the stimulation, no symbols or markers were used, due to the freedom given by the actual used stimulator to be independent from a strict cables' configuration.
A close-up view of one of the FES electrode connection sites is shown in Figure 16.

Figure 16: *UI of FES connection sites.*

3.6. Requirements Validation & Tests

A key challenge in designing smart pants equipped with EMG and FES electrodes is the assessment of the electrical and mechanical properties of the textile-based components, considering their intended long-term and daily use. This evaluation has been structured into two main phases:

- Initial characterization of the electrodes and conductive pathways, focusing on their baseline electrical performance;
- Durability testing, assessing the retention of these properties after prolonged simulated usage.

Initial Electrical Characterization

Drawing upon previous work [2], particular emphasis was placed on measuring the intrinsic impedance of the electrodes and associated conductive traces. Excessively high impedance can lead to power losses, local overheating, stimulation artifacts, and potential skin irritation.

The following steps are recommended:

- Verify the manufacturer's specified impedance using an LCR meter, both before and after integration into the garment, for both EMG and FES applications.
- Check the electrical continuity and conductivity of electrodes and traces using a multimeter, placing the probes at opposite ends of the conductive material.
- Assess the skin-electrode contact impedance by means of a potentiostat, applying a low-amplitude sinusoidal signal (about 10 mV) over a wide frequency range (0.1 Hz to 100 kHz). This enables the measurement of the complex impedance (magnitude $|Z|$ and phase θ), providing insight into contact quality and interface stability.

Durability and Resistance Evaluation

Given the intended autonomous and frequent use of the system, long-term durability becomes crucial. Therefore, a review of relevant testing protocols was conducted, with input from the FES research group at Politecnico di Milano, and based on Guzzardi's master degree thesis [33] and Rotzler et al. study [35].

The key tests for assessing textile electrode durability include:

- Mechanical fatigue tests, simulating repeated strain and bending;
- Washing tests, simulating exposure to thermal, chemical, and mechanical stress.

These procedures are summarized in Table 6 and 7 and should be performed prior to electrode integration into the garment.

Test Type	Technical Description
Stretching Test	Repeated elongation cycles (20–30% strain, about 1000 repetitions), simulating tensile stress during use. Assesses mechanical integrity and impedance variation.
Repeated Folding Test	Cyclic folding of the electrode (500–1000 cycles), simulating daily wear and manual washing. Evaluates trace fragility and resistance increases.

Table 6: *mechanical tests.*

Test Type	Technical Description
Soaking Test	Immersion in 30-40°C water with detergent for 60 minutes (5 consecutive cycles). Assesses the effects of thermal and chemical stress on impedance and resistance.
Washing Machine Test	Washing in a machine (10–100 cycles, 30 min each at 30-40°C and 400 rpm), followed by air drying. Evaluates changes in electrical properties after repeated mechanical, thermal, and chemical stress.

Table 7: *washability tests.*

In this study, practical testing was not performed due to the conceptual nature of the design. However, the selection of commercially available electrodes partially mitigates this limitation, as their durability and resistance are supported by the manufacturer's specifications.

Another important validation related to smart garments is the testing of user experience and usability. In the context of the project, this aspect mainly deals with the effectiveness and efficiency of the user interfaces, such as the choice of colors for the cables connections and the symbols adopted on the external cases of the control units for the correct placement. Standardized protocols and questionnaires (e.g. SUS survey) are the ideal solution for the evaluation of the system usability, with the possibility to retrieve also quantitative information.

3.7. Final Concept & Possible Challenges

The last section of the ideation phase will deal with the final concept and the analysis of possible criticalities that may arise. Now, all the parts previously analyzed will be summarized to describe how the final version of the pants will look like. Starting with the materials used, the choice was to use pants realized with a combination of polyamide and elastane because of its elasticity and adequate compression properties. Considering the electrodes, ElectroSkin GECKO were selected because of the possibility to use them for both EMG and FES and because of their washing resistance, cytotoxic, electrical and technical properties. Together with these electrodes, also Phantom Tape X and Phandom link were selected because of their durability and reduction of occupied space. Regarding the position of the electrodes, the pants will have bipolar electrodes for both EMG and FES for the rectus femoris, gluteus maximus, bicep femoris and gastrocnemius medialis of both the leg. This muscles will be alternatively stimulated to have a fluid cycle. The FES electrodes will be connected to already existing FES wires by snap buttons positioned as in Figure 12. The EMG instead, will be connected to three different control units for each leg as presented in the section 3.4. Moreover, the connection was thought to be as intuitive as possible with the use of colors and icons as already explained.
By looking at Figure 12, it is possible to notice the posterior part of the thigh is the zone with the highest density of Phantom Tape X due to the fact that in that region there is the presence of both the gluteus maximus

and the bicep femoris. As a consequence, this could be a critical zone for breathability. One of the most relevant articles related to this topic is Coull's one [38], who has reported a study concerning the difference of sweating considering several subjects of different ages at rest, but also during physical activity (walking on a treadmill). For the purposes of this project, the results obtained related to the sweat rate of the lower legs were analyzed. Figure 17 shows the areas of interest.

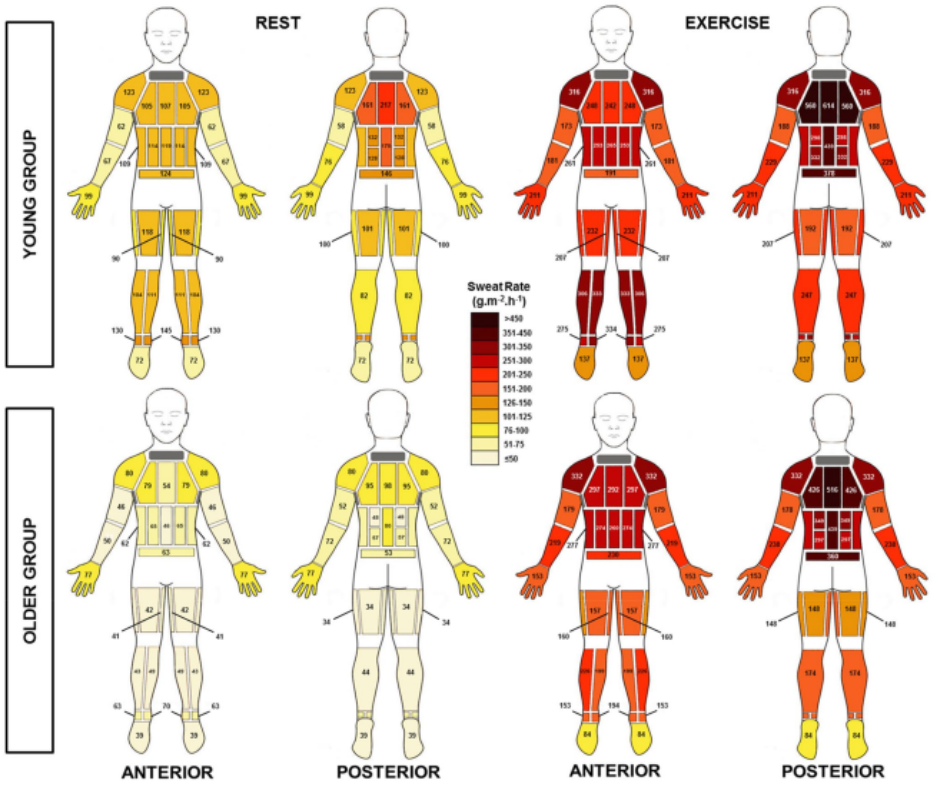

Fig. 5 Body maps of absolute regional median sweat rates (g m^{-2} h^{-1}) during the rest and exercise period (32 °C and 50% RH) in the young (18–30 years) and older (60–80 years) group

Figure 17: *sweat maps of young and elderly subjects at rest and during physical activity (walking on a treadmill).*

As it is possible to notice, the most critical region previously considered (the posterior part of the thigh) is characterized by a low value of sweat rate, the lowest one considering the legs. On the other hand, the gastrocnemius region, which is characterized by high sweat rates, has been designed with the smallest amount of Phantom Tape X possible. Eventually, the breathability will be certainly decreased by the presence of the tracks, but this reduction has been limited with proper design choices. Moreover, the introduction of zones with higher breathability in the regions where no cables are present could be a solution to counteract even more the effect of the parts subjected to thermo adhesion.

Focusing on the outer part of the pants, the placement of the FES attachment can be observed. The connections were placed externally to limit the intrusiveness of the FES cables and to be of easy access for the users. Indeed, even the gastrocnemius FES electrodes can be easily connected since they are at the knee level. Different colors have been selected according to the target muscle to guarantee an intuitive connection as explained in User-Interface section 3.5. Eventually, the EMG control units are easily detachable to guarantee the possibility to wash the garment. Figure 18 displays the final design of the ideated concept, showing both the outer side with the several connections, and the inner part with the focus on the electrodes and tracks placement.

Figure 18: *final concept design. Outer part (left) and inner part (right).*

4. Prototyping

In this section, the description of the realized prototype will be presented. Its aim was to test the detection of the muscular activity thanks to a EMG measurement in a context similar to the one of the final product. No testing or prototyping was conducted with the integrated FES electrodes. This decision was based on the extensive prior testing already performed on such electrodes, the need for expert supervision to ensure correct use of both the trike and the stimulation system, and, most importantly, the significant differences between the available electrodes and those selected for our concept design. The materials used were the ones available and not the ones described previously to realize the final product. An overview of the materials used is presented:

- *Decathlon leggings made of 92% of cotton and 8% of elastane.* This pants were used since they are very cheap, even though cotton is not the best solution for thermo adhesion;

Figure 19: *Decathlon leggings.*

– *Conductive fabric.* This fabric allowed the transmission of the signal acquired at the snap buttons.

Figure 20: *conductive fabric.*

– *Thermo adhesive film.* This film was important to guarantee the adhesion between the conductive material and the pants, and to guarantee the skin contact only in correspondence of the electrodes, while the remaining part of the conductive material should have been insulated from the skin to avoid the detection of other inputs different from the desired ones.

Figure 21: *thermo adhesive film.*

– *Steel snap buttons.* These buttons were used in order to guarantee the transmission of the signal from the textile material to the control unit (Myoware 2.0).

Figure 22: *snap button.*

– *Myoware 2.0.* This hardware is characterized by a triangular shape with a pin in each vertex. They are used for the two signals coming from the electrodes in bipolar configuration and the one coming from the reference electrode. Moreover, it is equipped with a potentiometer that allows the user to manually set a proper gain.

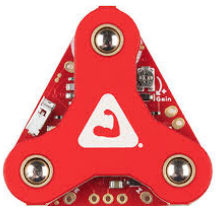

Figure 23: *Myoware 2.0.*

– *Elegoo wires (aluminium-copper).* These wires permitted the connection between the Myoware 2.0 and the Arduino Nano 33.

Figure 24: *Elegoo wires.*

– *Arduino Nano 33.* This control unit has been used to process the signals coming from the Myoware and guarantee their transmission to a PC for further analysis

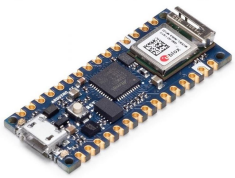

Figure 25: *Arduino Nano 33.*

Once all the materials used have been presented, the realization steps are explained. The prototype pants have been realized, but since the aim of this section is to test the feasibility of the muscular activity's extraction, only a single muscle was analyzed. More specifically, the rectus femoris of the left leg was the chosen one. The next points were followed for the realization:

1. *EMG position.* At first, the placement of the EMG sensor of the rectus femoris was located based on the indication provided by the SENIAM. Accordingly, the location was defined at 50% on the line from the anterior spina iliaca superior to the superior part of the patella;
2. *Designing of the electrodes and thermo adhesive film.* The electrodes were designed to have an inter-electrode distance of 20 mm (indication by SENIAM). In this step also the shape of the thermo adhesive film was designed to permit the conduction in the correct position and to guarantee the insulation in the remaining zones. Finally, a proper design and dimensioning of the components is required to correctly realize the successive thermo adhesion step.

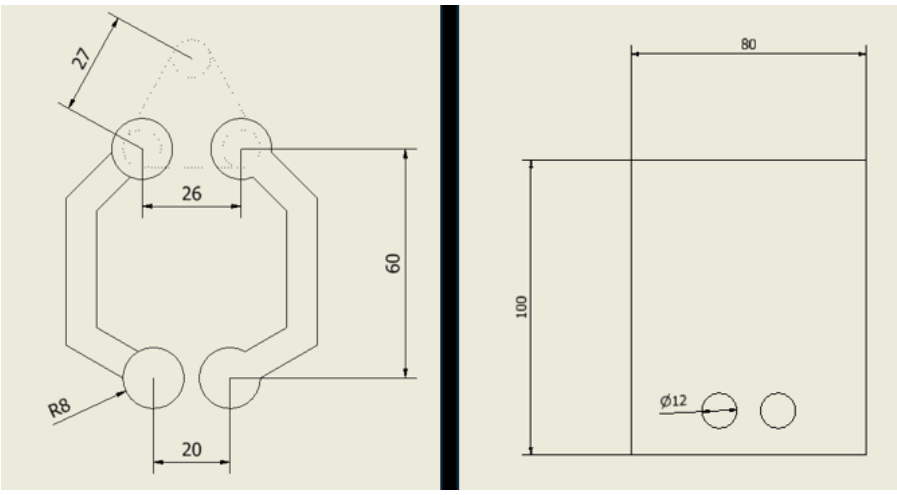

Figure 26: *electrodes (left) and thermo adhesive film (right) design and modeling.*

3. *Cut of the electrodes and thermo adhesive film.* At this point, both the electrodes and the thermo adhesive film have been cut through laser cutting.
4. *Thermo Adhesion.* The electrodes and the thermo adhesive film were carefully placed on the machine to correctly match the holes on the film with the recording part of the electrodes. In this step a part of the snap buttons was also inserted in order to facilitate connection to the other part once the adhesion was completed. The process was made at 150°C for 25 seconds with a thermo adhesive press.
5. *Close of the snap buttons.* After the thermo adhesion, the snap buttons were closed and the pants were ready to be connected to the Myoware to start the analysis.
6. *Conductivity test.* Before the analysis, the conductivity was measured with a multimeter be sure it correctly worked, otherwise the acquired signals would have been too noisy.

An important consideration to highlight is that the thermo adhesion was made only on the electrodes of the bipolar configuration and not on the reference one. Indeed, it was realized with a portion of conductive material and a snap buttons to connect it with the Myoware. The electrode was connected to the snap button by means of a previously welded cable.
To make the acquisitions, the three snap buttons were connected to the correspondent location in order to get the EMG signal. At this stage, the connection to the Arduino and consequently to the PC was realized too. Before acquiring the data, the gain was tuned by means of a screwdriver in order to properly retrieve the signal. The overall setup is presented in Figure 27.

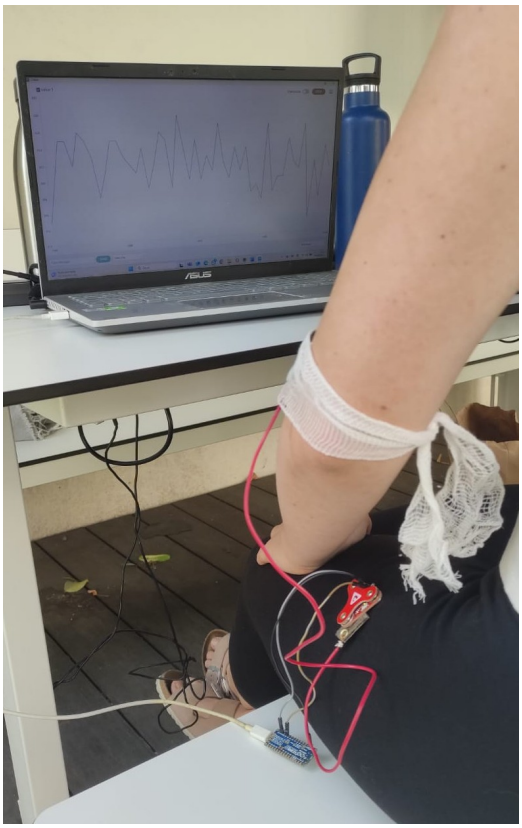

Figure 27: *overall setup.*

The acquisition consisted of 30 seconds of recording. During this period, the subject was asked to perform two prolonged contractions, where the first one was isometric. The acquisitions were saved and then analyzed with MATLAB. An example is reported in Figure 28.

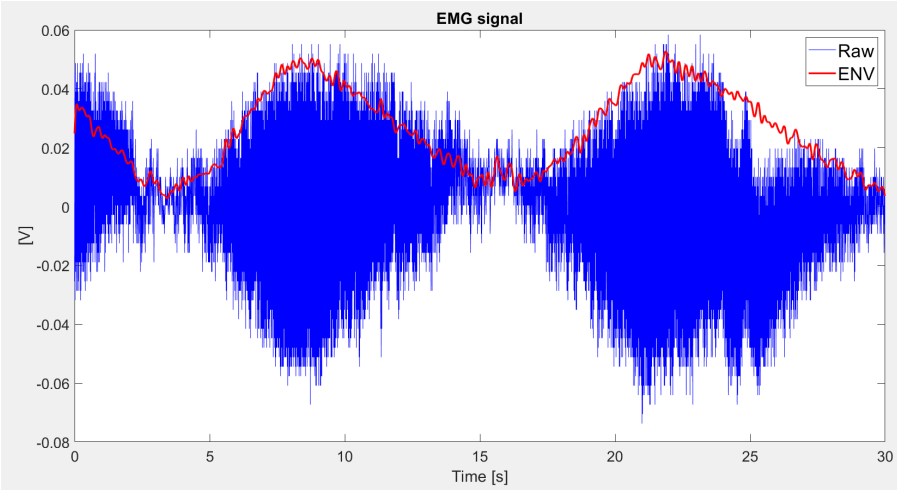

Figure 28: *EMG extracted raw signal and computed envelope.*

As it is possible to notice, the two contractions are quite recognizable in the graph, even considering the raw signal. The first contraction starts at 5 seconds and lasts for about 10 seconds, while the second one starts at around 18 seconds and lasts until 26 seconds more or less. This is in accordance to the instructions given to the subject during the test. Focusing on the amplitude of the signal, the values are between 40-50 mV, that is quite high as an EMG amplitude. However, this could be due to the gain that was manually tuned and also to the placement of the reference electrode, which was fixed by means of a gauze to the elbow (bony part not involved in the movement). Nevertheless, the gauze couldn't guarantee a uniform pressure and this may have caused a decrease in the Signal to Noise ratio. To get rid of this effect, during the computation of the envelope, a low pass filter at 5 Hz has been used. The envelope is the result of the amplified and rectified signal.
Another issue related to the setup realization was the sampling frequency used. For EMG recordings the acquisition frequency is usually a least 1000 Hz, however, this would have been translated only to too few seconds of acquisitions due to Arduino's limited memory. As a consequence, the frequency was decreased up to 100 Hz in order to acquire at least two contractions. This was a big limitation considering the final aim of the product, leading to the research of different solutions for the final concept.
In conclusion, even though the setup was not optimal for the acquisition because of all the problems mentioned before, the system was able to detect the two contractions and to show preliminarily the functions expected by the concept design. Most of the problems should be solved by using all the materials and techniques presented above, and the final garment should be able to detect higher quality data compared to this prototype.

5. Conclusions

The aim of this report was to develop a proof of concept of pants which incorporates both EMG and FES electrodes in order to enhance an autonomous use of the trike in the context of Active^3 project. Currently, few garments having both FES and EMG electrodes are commercially available, giving the product the possibility to fill the gap. However, the solution proposed stimulates and records the target muscles (rectus femoris, gluteus maximus, bicep femoris and gastrocnemius medialis) in different occasions: training with FES and evaluating the results with EMG electrodes by monitoring the fatigue of the user is the final scenario. The fatigue analysis could done by computing the RMS value of the muscular activity over time to see the improvements made between different training sessions. In the prototyping previously proposed, the fatigue analysis was not performed due to the low amount of data available. Future developments may include more standardized protocols to evaluate the fatigue and eventually also the possibility to use FES and EMG electrodes at the same time. In the literature, few solutions able to do that are present. However, since the aim is to evaluate the improvements in physical health by looking at the fatigue in an autonomous training session, there is no need to have both the systems active at the same time.

EMG control units are equipped also with IMUs to track the kinematic of the cycling in order to properly set the FES parameters. Future studies could delve into the specific development of the control units to embed all the required components, while also investigating the kinematic parameters hypothetically obtained by the integrated inertial measurement units.

Both the electrodes typologies are realized with already available technologies: GECKO ElectroSkin, PhantomLink and PhantomTape X are used. This permits to have low level of impedance, good elasticity and washability properties, essential for a wearable system.

The pants design is thought to be as more intuitive as possible. Colors and shapes guide the user in the attachment of FES wires and EMG control units, while the snap buttons guarantee a fast and safe attach.

The materials selected for the pants (polyamide and elastane), together with a warp knitting technology, guarantee to have satisfying compression properties to enhance good acquisitions, together with the possibility to perform thermo adhesion and good resistance to damages.

Eventually the sweat problem has been observed since thermo adhesion may compromise the pants brethability. The parts subjected to this technique are most likely placed on body zones characterized by a low sweat rate. In addition, parts with higher breathability properties are realized where there is no thermoadesion on the pants. In this way the reduction in breathability is limited.

To test the feasibility of the solution, a prototype has been made. It demonstrated the possibility to quite easily collect muscular activity, however, due to the limitations of the devices used in this stage, a more appropriate solution should be produced. This would involve the use of the chosen material to test the SNR and the quality of the signals acquired, together with a more proper evaluation and dedicated protocol development of the fatigue. Indeed, the current setup permits only few seconds of recording with a sampling frequency of 1000 Hz, that is the proper one for EMG acquisitions.

Eventually, all the study has been conducted on L-size pants, different users may need other sizes. In this context, the scaling process should be carefully defined. It has been proposed to scale the position of the electrodes with percentages computed considering the coordinates of the electrodes with respect to the total length or half width of the pants for the vertical and horizontal positions respectively. This process is limited since in the actual study the positions are defined referring only to a single male 23 years old subject. To account the inter-subject variability, more users should be involved in the study in order to compute mean positions on different-size pants.

6. Our Product: Name, Values & Logo

Name - "ReActive"

The name ReActive carries layered meanings that reflect the mission of the project. The word "Active" evokes motion, vitality, and the aspiration to maintain a dynamic lifestyle throughout aging (while also directly referencing the previously mentioned Active[3] project). The prefix "Re-" introduces the idea of reactivity, renewal, and proactive response, highlighting the technology's capacity to stimulate and support movement. Simultaneously, our technology offers active care and strength, guiding and empowering movement in later life.

Values

ReActive is built upon a core set of values that inform every design and technological choice:

- *Care & Protection:* this value is expressed through the device's ability to support users with gentle yet effective action, promoting healthy and active aging.
- *Non-invasive Technology:* thanks to integrated surface EMG and FES electrodes, users experience efficient neuromuscular support without discomfort or intrusive hardware.
- *Accessibility & Autonomy:* the product's ergonomic design, comfortable fabrics, and user-friendly interface are tailored to the cognitive and physical needs of older adults, restoring independence and confidence.
- *Simple Elegance:* the aesthetic is clean and understated, yet refined - designed to communicate technological quality while remaining familiar and reassuring. Design becomes an integral part of the well-being experience.

Logo

Figure 29: *ReActive logo.*

The ReActive logo is a visual synthesis of the project's values and technological components. At its core lies a stylized waveform, reminiscent of EMG signals - capturing and stimulating muscle activity. This wave is enclosed within a protective circular shape, topped by a leaf, symbolizing vitality, nature, and sustainable, harmonious activity. The overall design suggests a dynamic embrace between technology and nature, between innovation and care. It is a simple yet powerful symbol, designed to be instantly recognizable and to convey trust, innovation, and well-being. The logo stands as the emblem of active ageing promoted by ReActive: a vision where smart design, accessibility, comfort and aesthetic sensibility converge.

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